

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1 AT 1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
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Date:	22-07-2026	Duration:	60 minutes

Code of Conduct

I Jeyakannan /192525376 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: jeyakannan.

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

Problem Statement: Fraud Detection and Financial Transaction Monitoring System

Domain : Bank and cyber

1.Understand the Problem Statement

1.1 Explain the Problem in Your Own Words

The **Fraud Detection and Financial Transaction Monitoring System** is developed to identify and prevent fraudulent financial activities before they result in financial loss. With the rapid growth of online banking and digital payment platforms, fraudulent transactions have become increasingly sophisticated. Traditional monitoring methods are often slow and less effective in detecting new fraud patterns. This proposed system continuously monitors transactions, analyzes user behavior, identifies suspicious activities, and generates instant alerts to improve financial security, customer trust, and regulatory compliance.

1.2 Identify the Objectives of the Proposed System

The primary objective of the proposed system is to improve the security and reliability of financial transactions through intelligent fraud detection. The system aims to monitor transactions continuously while minimizing financial risks and operational delays.

Objectives:

- Detect fraudulent transactions in real time.
- Reduce financial losses and false alerts.
- Improve transaction monitoring efficiency.
- Protect customer financial information.
- Support regulatory compliance.
- Increase customer confidence in digital banking services.

By achieving these objectives, the system creates a secure and efficient financial transaction environment.

1.3 State the Scope of the Problem

The scope of the system includes monitoring transactions performed through online banking, mobile banking, ATMs, debit cards, credit cards, and digital wallets. It analyzes transaction history, customer behavior, device information, and transaction locations to identify suspicious activities. The system is suitable for banks, financial institutions, and payment service providers. It focuses on fraud detection, transaction monitoring, alert generation, reporting, and secure data management while ensuring scalability, reliability, and compliance with financial regulations.



2. Identify Key Stakeholders and Challenges

2.1 Identify All the Stakeholders Involved in the System

The success of the proposed system depends on the participation of several stakeholders who contribute to its development, maintenance, and operation.

Key Stakeholders:

- Bank Customers
- Bank Employees
- Fraud Investigation Team
- System Administrators
- Financial Institutions
- Payment Gateway Providers
- Regulatory Authorities
- Software Developers

2.2 Explain the Roles and Responsibilities of Each Stakeholder

Every stakeholder has specific responsibilities that contribute to the efficient functioning of the fraud detection system.

- **Customers:** Perform secure financial transactions and report suspicious activities.
- **Bank Employees:** Verify customer information and respond to fraud alerts.
- **Fraud Analysts:** Investigate suspicious transactions and prevent financial losses.
- **System Administrators:** Maintain servers, databases, and system security.
- **Software Developers:** Design, develop, and upgrade the software.
- **Financial Institutions:** Establish fraud prevention policies.
- **Regulatory Authorities:** Ensure compliance with financial regulations.
- **Payment Gateway Providers:** Process secure online transactions.

2.3 Discuss the Major Challenges and Issues Faced by the Stakeholders

Financial stakeholders face several challenges due to the continuous evolution of cyber threats and fraudulent activities.

Major Challenges:

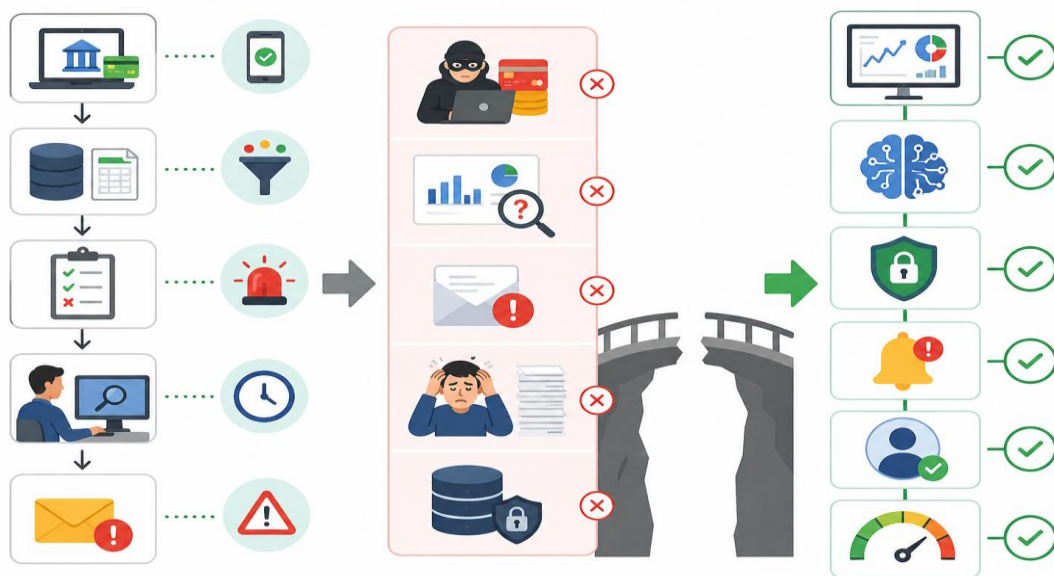
- Increasing cyber fraud and identity theft.

- High volume of daily financial transactions.
- False-positive fraud alerts.
- Delayed fraud investigations.
- Maintaining customer privacy and data security.

3. Analyze the Current Approach

3.1 Describe the Existing System or Current Process

Most financial institutions currently use rule-based fraud detection systems that compare transactions against predefined conditions. When unusual activity is detected, alerts are generated for manual verification by fraud analysts. Although this approach can detect common fraud patterns, it often struggles to identify new or sophisticated attacks. Manual investigation increases response time and operational costs, making the existing system less efficient for handling millions of transactions processed every day.



3.2 Identify Its Strengths and Limitations

The current system provides basic fraud detection capabilities but has several operational limitations.

Strengths

- Reliable transaction recording.

- Easy implementation and maintenance.
- Supports regulatory compliance.
- Detects common fraud patterns.

Limitations

- Cannot identify newly emerging fraud techniques.
- High false-positive rate.
- Heavy dependence on manual verification.
- Limited scalability for large transaction volumes.

Improving these limitations is essential for building a smarter fraud detection system.

3.3 Analyze the Problems or Gaps That Need to Be Addressed

Several gaps exist in the existing fraud detection approach that reduce its overall effectiveness.

Problems Identified:

- Delayed fraud detection.
- Limited real-time monitoring.
- Poor adaptability to new fraud patterns.
- High operational costs.
- Excessive false alerts.
- Lack of intelligent learning capabilities.

The proposed system addresses these issues by introducing automated monitoring, intelligent analysis, and faster fraud detection mechanisms.

4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a Software-Based Solution for the Given Problem

The proposed software solution is an intelligent fraud detection system that continuously monitors financial transactions and analyzes customer behavior using advanced analytical techniques. Whenever suspicious activities are identified, the system automatically generates alerts for investigation. A centralized dashboard allows administrators to monitor transactions efficiently while maintaining secure authentication, encrypted communication, and protected financial data. This solution improves fraud detection accuracy, operational efficiency, and customer trust through continuous real-time monitoring.

4.2 Describe the Major Functional Modules of the Proposed System

The proposed system is divided into independent modules to improve maintainability and performance.

Major Functional Modules

- User Authentication
- Transaction Monitoring
- Fraud Detection Engine
- Alert Management
- Investigation Dashboard
- Report Generation
- Customer Notification

- System Administration

Each module performs a dedicated function while working together to provide an efficient, secure, and reliable fraud detection system.

4.3 Identify the Functional and Non-Functional Requirements

The system should satisfy both functional and non-functional requirements to ensure successful implementation.

Functional Requirements

- User Registration and Login
- Transaction Monitoring
- Fraud Detection
- Alert Generation
- Report Generation
- Customer Notification
- Administrator Management

Non-Functional Requirements

- Security
- Reliability
- Scalability
- Maintainability
- High Performance
- Usability

- Data Privacy
- Availability

These requirements ensure secure and efficient financial transaction monitoring.

4.4 Explain How Software Engineering Principles Are Incorporated into Your Solution

The proposed solution follows important software engineering principles to improve overall software quality.

- **Modularity:** Separate functional modules simplify development and maintenance.
- **Scalability:** Supports increasing transaction volumes efficiently.
- **Maintainability:** Allows easy software updates and feature enhancements.
- **Reliability:** Ensures consistent fraud detection with minimum downtime.
- **Usability:** Provides an intuitive interface for users and administrators.
- **Security:** Protects sensitive financial information using authentication and encryption.

These principles ensure that the software remains efficient, secure, and easy to maintain.



5. Justify the Chosen Methodology/Framework

5.1 Select an Appropriate SDLC Model

The **Agile Software Development Methodology** is the most suitable approach for developing the Fraud Detection and Financial Transaction Monitoring System. Agile supports iterative development, continuous testing, customer feedback, and rapid software updates. Since fraud techniques evolve frequently, Agile enables developers to improve detection algorithms regularly while delivering high-quality software. It also promotes collaboration among team members, making the development process flexible, efficient, and adaptable to changing business requirements.

5.2 Justify Why the Selected Methodology Is Suitable

Agile provides flexibility and continuous improvement, making it ideal for financial security applications.

Reasons for Selecting Agile

- Supports continuous software improvements.
- Quickly adapts to new fraud techniques.
- Encourages customer feedback.
- Reduces development risks.
- Enables frequent software testing.
- Delivers high-quality software faster.

These advantages make Agile an excellent choice for fraud detection system development.

5.3 Explain How It Supports Successful Project Development and Maintenance

Agile divides the project into smaller iterations, allowing continuous development, testing, and evaluation. Regular customer feedback helps developers identify problems early and improve software quality. Frequent software releases simplify maintenance by allowing new security updates and features to be added without disrupting existing operations. Agile also promotes teamwork, transparency, and faster issue resolution, ensuring the fraud detection system remains reliable, scalable, and effective throughout its operational lifecycle.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the Expected Benefits of the Proposed Solution

The proposed system provides several advantages for financial organizations and customers.

Expected Benefits

- Faster fraud detection.
- Improved transaction security.
- Reduced financial losses.
- Real-time monitoring.
- Better customer satisfaction.
- Improved regulatory compliance.
- Reduced manual investigation effort.

Overall, the solution enhances financial security while improving operational efficiency and customer trust.

6.2 Explain How the Solution Addresses the Identified Challenges

The proposed solution continuously monitors financial transactions and analyzes customer behavior to detect suspicious activities immediately. Automated fraud detection reduces investigation time and minimizes human errors. Strong authentication, encryption, and access control improve data security and customer privacy. The scalable architecture supports large transaction volumes, while continuous software updates allow the system to adapt to emerging fraud techniques, ensuring reliable fraud detection and improved financial protection.

6.3 Discuss Potential Risks, Implementation Challenges, and Limitations

Although the proposed system offers many advantages, certain challenges may arise during implementation.

Potential Challenges

- False-positive fraud alerts.
- High implementation costs.
- Data privacy concerns.

- Integration with existing banking systems.
- Continuous maintenance requirements.
- Frequent updates for evolving fraud techniques.

Proper planning, regular maintenance, and strong cybersecurity practices can minimize these risks and improve overall system performance.

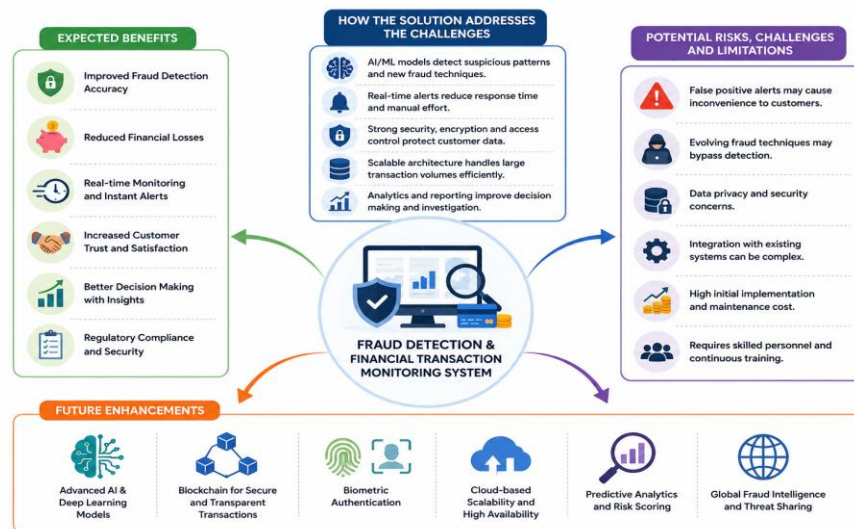
6.4 Suggest Possible Future Enhancements

The proposed system can be enhanced by integrating advanced technologies that improve fraud detection accuracy and system performance.

Future Enhancements

- Artificial Intelligence and Deep Learning.
- Blockchain-based transaction verification.
- Biometric authentication.
- Predictive fraud analytics.
- Cloud-based deployment.
- Real-time risk scoring.
- Global fraud intelligence integration.

These future improvements will make the system more intelligent, secure, scalable, and capable of handling future financial security challenges effectively.



Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5